

Development of a Multimodal IoT Network for Contactless Vital Sign Monitoring and Fall Detection

Problem Statement

The **rapid** growth of the aging population in industrialized countries and the **insufficient** support for safe **aging in place** create a need to develop **passive, scalable technologies** enabling comprehensive **health monitoring** and **risk detection**.

Existing Approaches

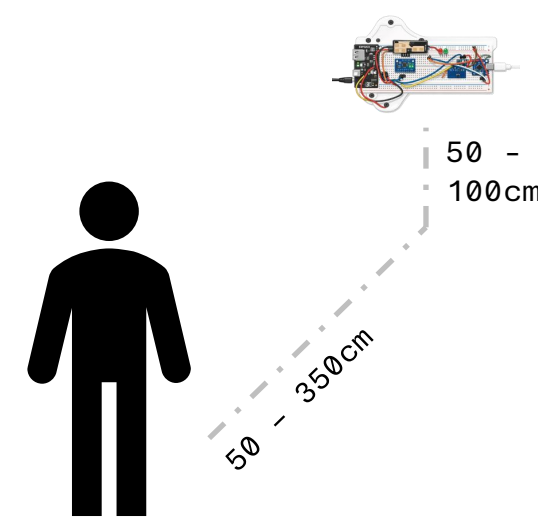
Wearable Sensors	- Generally accurate motion and vitals Poor compliance, discomfort, maintenance, high false positives
Vision-Based Systems	- Rich spatial information Privacy concerns, high compute requirements, occlusion sensitivity
mmWave Radar	- Contactless, privacy-preserving - Sensitive to environment, motion ambiguity, lacks context
Vibration / Floor Sensors	- Captures impacts, gait signatures False positives, installation difficulty, ambiguous events

This work combines mmWave radar and piezoelectric vibration sensing to leverage **complementary strengths** and reduce **single-modality failure** modes. (⚡) + (📡)

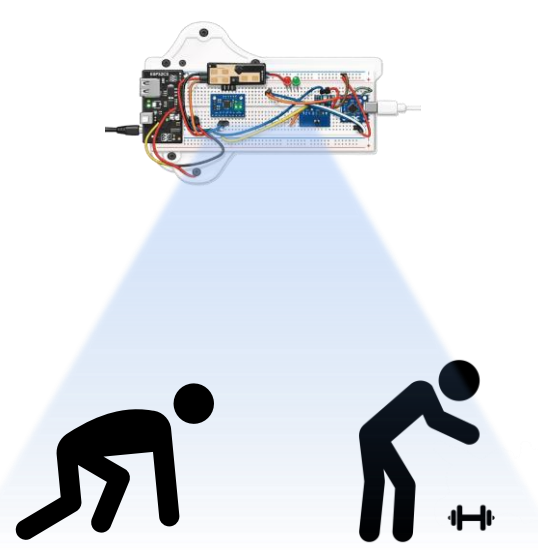
Data Collection Methodology

Anomalous Activity Detection

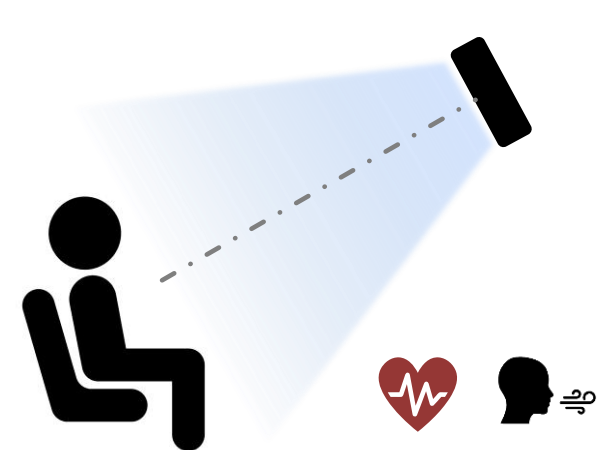
- Sensor node is wall-mounted, continuous time-series data logging at **20Hz**
- Subject clothing, sensor height, and room setup are varied to **diversify** environment multipath effects and radar reflection



- Collecting large samples of actual falls is **unrealistic** and **ethically** dubious
- Focus on **anomaly detection**: most recorded data is **normal** ADLs (*walking, sitting, standing at varying distances, speeds, directions*)



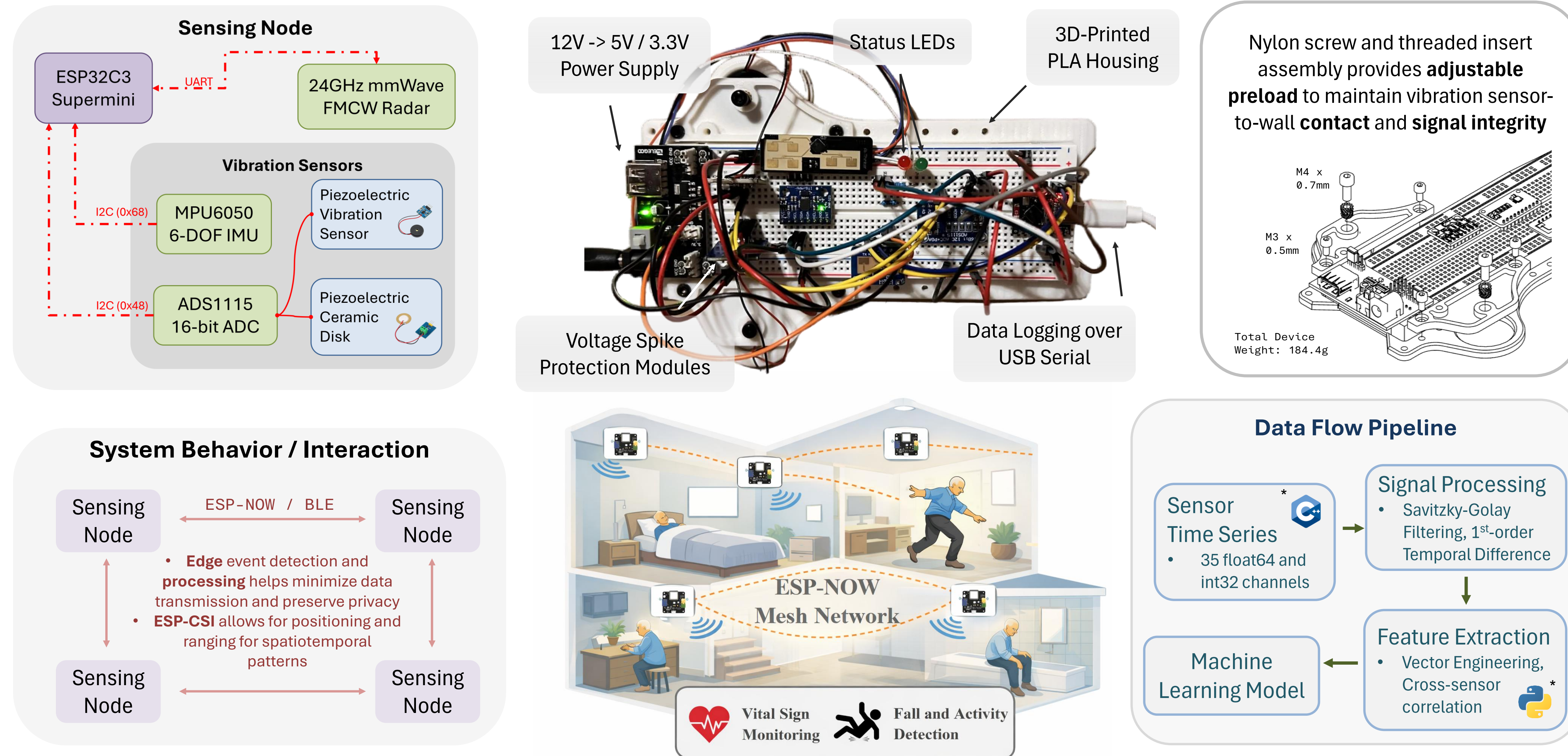
- Simulated **anomalous** activities are recorded (*staged falls, stumbles, limp*)
- “Confuser” events known to trigger **false positives** in individual sensors (*heavy object drops, standing suddenly*)



Vital Sign Monitoring

- Subject sits stationary, approximately 100cm between chest and sensor node
- Respiration rate** and **heart rate** are recorded with a Huawei Band 4 Pro-70E

System Overview



Vital Sign Monitoring

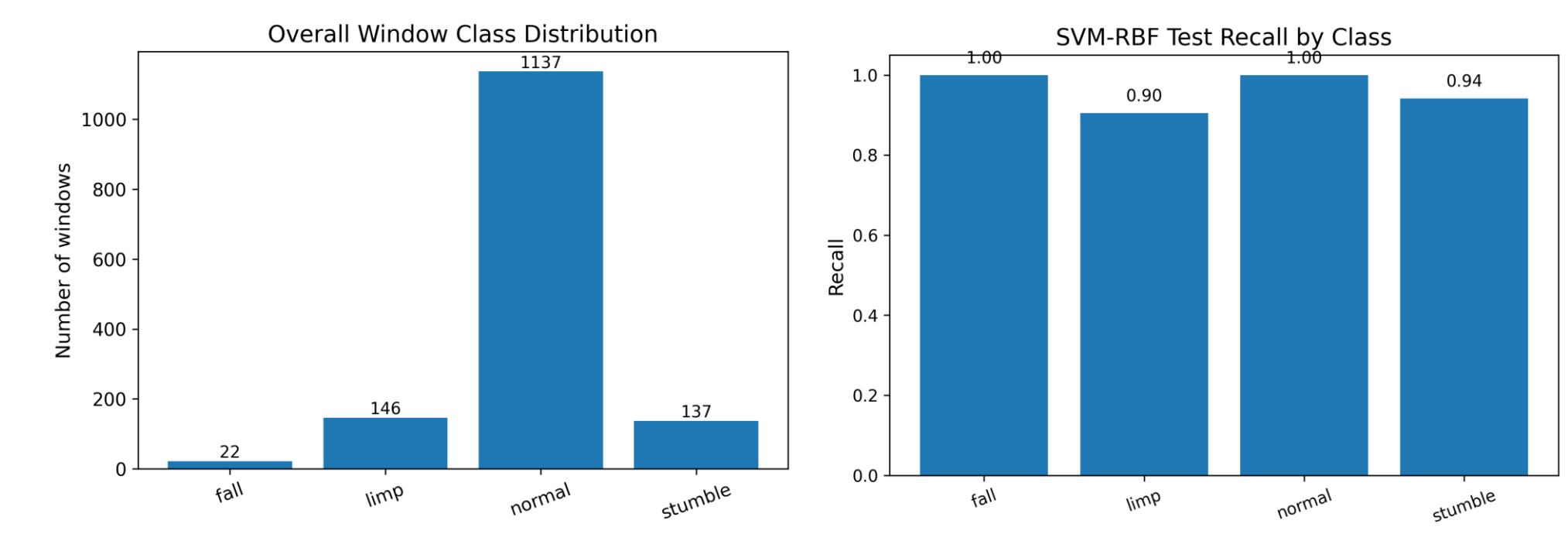


Radar signals **rise/fall** in conjunction with chest **expansion and contraction** when inhaling (and exhaling)

Cross-correlation and temporal feature calculations are applied between individual radar signals at high sensitivity **Peak analysis** is performed on **Savitzky-Golay** filtered signals, closely matching (<3% error) true **respiration rate** at short distances

The low-cost 24GHz radar modules used in this work do **not** provide raw quadrature radar data. Unfortunately, attempts at **physics-based** heartrate sensing and calculation have been so far **unsuccessful**.

Conclusions



- Multimodal** mmWave + vibration features enable **strong** multiclass anomaly discrimination using classical CPU-trainable models, even with heavy dataset **class imbalance** for anomalous features.
- Cross-sensor** correlation and lag features were effective for representing coupled motion-impact behavior.

Future Work

- Integrate WiFi Channel State Information as additional **sensing modality** for pose estimation, breathing rate, and heartbeat
- Train and evaluate **deep learning** (CNN) model performance for more **generalizable** fall detection
- Use spatiotemporal information to control node activation/sleep, optimizing **power consumption**

Key References

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Full Bibliography available in Research Notebook.

The student researcher would like to acknowledge Dr. Christa Hughes and Ms. Mina Bradley for their invaluable support. He would also like to thank Mr. Hongbiao Zhao and Mrs. Minfang Wu for always pushing him to do his best and for putting up with his late-night procrastination.

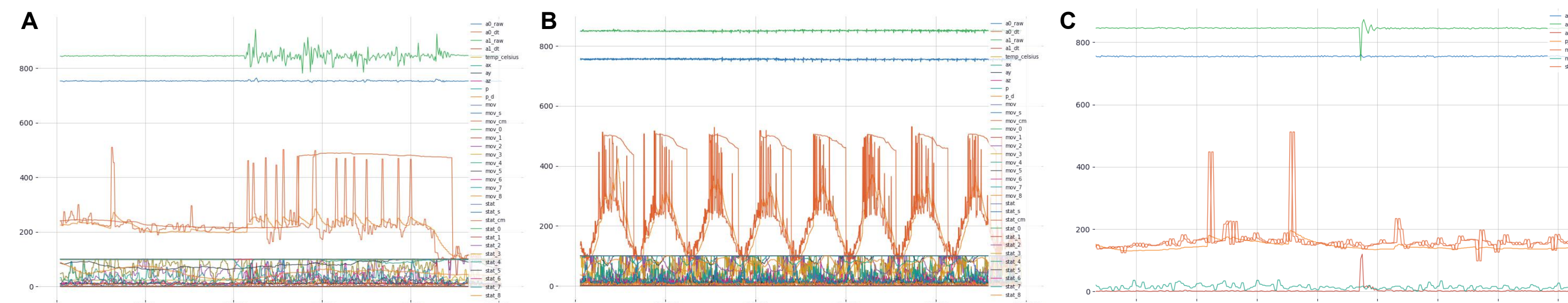
Fall Detection Models

Dataset

- Time series sensor data .csv files are sampled at **20Hz** and recorded for durations varying from ~25.0s to ~120.0s
- Files are split into **4.0s** (80 timesteps) **windows** with 50% overlap, then split by file into **training, testing, and validation**

Classification	# Files	# Windows
Normal	49	1137
Stumble	22	147
Limp	11	152
Fall	14	32

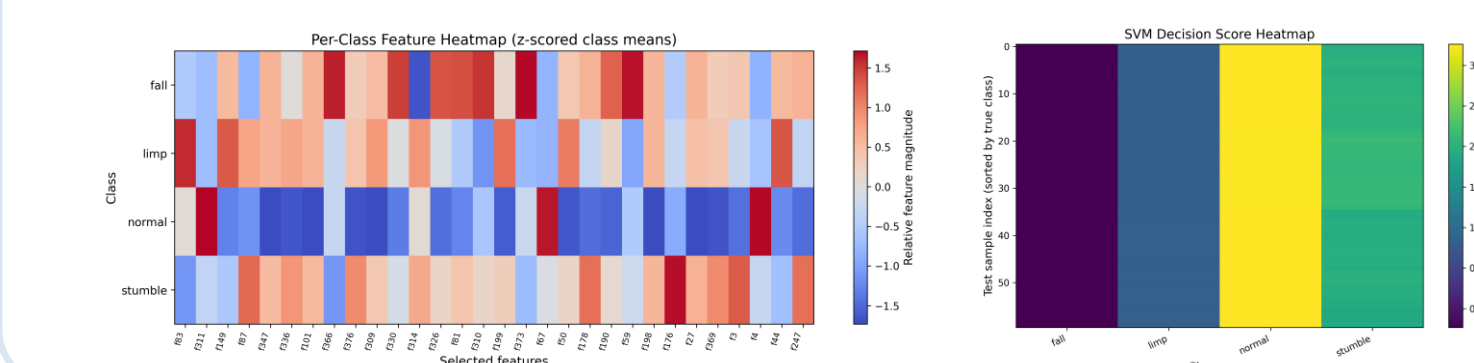
Example Signal Plots



Multiclass Classification Models

Support Vector Machine (SVM-RBF): Radial Basis Function

Balanced Accuracy	f1_macro	f1_weighted
0.9614845	0.9602787	0.9860756



Random Forest Classification (RF)

Balanced Accuracy	f1_macro	f1_weighted
0.8842787	0.9047619	0.9767442

Hyperparameters:
320 trees, max_depth=12, min_samples_split=8
min_samples_leaf=4, max_features="sqrt"
class_weight="balanced_subsample"

